

LESSON 5-1 • SESSION 1 • PRACTICE • FORM A

Area of Parallelograms

Learning target – I can find the area of a parallelogram by multiplying the base by the perpendicular height. · 6.G.A.1

REMEMBER

- Area of a parallelogram: **$A = b \times h$** .
- The **height** makes a square corner with the base. The slanted side is not the height.
- Missing measure: divide the area by the one you know ($90 \div 18 = 5$). Label **square units**.

WORKED EXAMPLE

Base 9 in., height 4 in., slanted side 5 in. Find the area.

Step 1 Base = 9 in., height = 4 in.

Step 2 Skip the 5 in. slanted side.

Step 3 $A = 9 \times 4 = 36$

Answer **36 in²**

WORDS **parallelogram** (*paralelogramo*) · **base** (*base*) · **height** (*altura*) · **perpendicular** (*perpendicular*)

PRACTICE

show your work · label units

1. Find the area of each parallelogram with base b and height h .

a. $b = 18$ cm, $h = 7$ cm _____

b. $b = 12$ in, $h = 6$ in _____

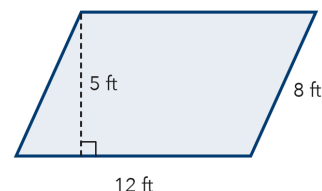
c. $b = 11$ ft, $h = 7$ ft _____

d. $b = 25$ m, $h = 4$ m _____

2. **School Banner.** A banner is a parallelogram.

a. Find the area. _____ ft²

b. Why is the 8 ft side not used?



3. **Floor Tile.** A tile has a base of 15 cm and a height of 10 cm.

a. Area of one tile _____ cm²

b. Area of 3 tiles _____ cm²

4. **Missing Height.** A sign has an area of 90 in² and a base of 18 in. Find its height.

Hint What times 18 makes 90?

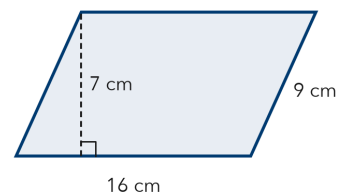
Answer _____ in.

5. **Paper Strip.** Find the area of the paper strip.

Hint Use the height with the square corner, not 9 cm.

Show your work.

Answer _____ cm²



6. **Window Pane.** A window pane is a parallelogram. Its area is 96 in² and its base is 8 in. Find the height.

Hint Divide the area by the base.

Answer _____ in.

- 7. Two Panels.** Panel X has base 10 cm and height 13 cm. Panel Y has base 26 cm and height 5 cm. Find each area. Which covers more space?

Hint Find both areas, then compare.

Show your work.

Areas _____

- 8. Hallway Mural.** A mural is a parallelogram with a base of 15 ft and a height of 8 ft. One can of paint covers 50 ft².
- a.** Find the area. _____ ft²
 - b.** How many cans are needed? Explain. _____
 - c.** A second mural has the same height and double the base. Find its area. _____ ft²

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Marisol’s Work.** A parallelogram has base 20 cm, slanted side 17 cm, and height 15 cm. Marisol wrote: “ $A = 20 \times 17 = 340 \text{ cm}^2$.” Explain her mistake. Then find the area.

TEST PRACTICE

choose the best answer

- 10.** A parallelogram has a base of 15 m and a height of 6 m. What is its area?
- A** 21 m² **B** 42 m² **C** 90 m² **D** 180 m²
- 11.** A parallelogram has an area of 72 cm² and a height of 8 cm. What is its base?
- A** 9 cm **B** 64 cm **C** 80 cm **D** 576 cm

SELF-CHECK CODE

find your answer · copy its letter into that problem’s box · spell a math word

12 R	144 X	112 A	96 Z	120 S	150 Q	60 S	5 U	130 E	768 J
#2a		#3a	#4	#5	#6	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____